

ACM-ICPC TEAM REFERENCE DOCUMENT

Skill++ (Gogolev, Drachev)

Contents

1	General	1
1.1	C++ Template	1
1.2	Compilation	1
1.3	Check	1
2	Math	2
2.1	Simpson	2

1 General

1.1 C++ Template

```
#pragma GCC optimize("O3,unroll-loops")
#define _CRT_SECURE_NO_WARNINGS
#define ln '\n'
#define All(x) (x).begin(), (x).end()
#define Allr(x) (x).rbegin(), (x).rend()
#define Allf(x) x.begin() + 1, x.end()
#define fi(a, b) for (auto i = (a); i <= (b); i++)
#define fj(a, b) for (auto j = (a); j <= (b); j++)
#define fo(a, b) for (auto o = (a); o <= (b); o++)
#define fq(a, b) for (auto q = (a); q <= (b); q++)
#define fe(a, b) for (auto e = (a); e <= (b); e++)
#define fw(a, b) for (auto w = (a); w <= (b); w++)
#define fdi(a, b) for (auto i = (a); i >= (b); i--)
#define fdj(a, b) for (auto j = (a); j >= (b); j--)
#define fdo(a, b) for (auto o = (a); o >= (b); o--)
#define fdq(a, b) for (auto q = (a); q >= (b); q--)
#define fde(a, b) for (auto e = (a); e >= (b); e--)
#define fdw(a, b) for (auto w = (a); w >= (b); w--)
#define fx(A) for (auto &x : (A))
#define fy(A) for (auto &y : (A))
#define fz(A) for (auto &z : (A))
#define pb push_back
#define mp make_pair
#define ins insert
#define sz(x) (int)x.size()
#include <string>
#include <vector>
#include <list>
#include <map>
#include <set>
#include <deque>
#include <queue>
#include <stack>
#include <bitset>
#include <algorithm>
#include <sstream>
#include <iostream>
#include <iomanip>
#include <cstdio>
#include <cstdlib>
#include <ctime>
#include <cstring>
#include <cmath>
#include <cstdlib>
#include <cassert>
#include <ctime>
#include <tuple>
#include <unordered_set>
#include <unordered_map>
#include <random>
#include <chrono>
#include <complex>
using namespace std;
#ifdef _DEBUG
#define LOCAL
#endif
mt19937_64 rnd(chrono::steady_clock::now().time_since_epoch().count());
```

```
typedef long long ll;
typedef unsigned long long ull;
typedef long double ld;
typedef vector<bool> vb;
typedef vector<int> vi;
typedef vector<ll> vll;
typedef vector<ld> vld;
typedef vector<char> vc;
typedef vector<string> vs;
typedef vector<vb> vvb;
typedef vector<vi> vvi;
typedef vector<vll> vvll;
typedef vector<vld> vvld;
typedef pair<int, int> pii;
typedef pair<ll, ll> pll;
template<typename A, typename B>
ostream& operator<<(ostream& os, pair<A, B> p) {
    os << "(" << p.first << ", " << p.second << ")";
    return os;
}
template<typename T>
ostream& operator<<(ostream& os, vector<T> v) {
    fi(0, sz(v) - 1) {
        os << v[i] << " ";
    }
    return os;
}
template<typename T>
ostream& operator<<(ostream& os, set<T> t) {
    for (auto z : t) {
        os << z << " ";
    }
    return os;
}
template<typename T1, typename T2>
ostream& operator<<(ostream& os, map<T1, T2> t) {
    cerr << endl;
    for (auto z : t) {
        os << "\t" << z.first << "→" << z.second << endl;
    }
    return os;
}
#ifdef LOCAL
#define dbg(x) {cerr << __LINE__ << "\t" << #x << " : ";
    << (x) << endl;}
#else
#define dbg(x) {}
#endif
#ifdef LOCAL
#define ass(x) if (!(x)) { cerr << __LINE__ << "\tassertion_
    failed: " << #x << endl, abort(); }
#else
#define ass(x) assert(x)
#endif
void solve() {

}

int main()
{
    ios::sync_with_stdio(0);
    cin.tie(0);
    cout.tie(0);
    cout << setprecision(20);
    #ifdef LOCAL
        freopen("input.txt", "r", stdin);
        freopen("output.txt", "w", stdout);
    #endif
    solve();
}
```

1.2 Compilation

```
#far association
g++ --std=c++17 -! -DLOCAL -O2 -Wall -Wl,--stack
=67108864 -o !.exe
```

1.3 Check

```
:start
gen.exe
sol.exe
copy output.txt answer.txt
NAME.exe
fc output.txt answer.txt
if ERRORLEVEL 1 goto end
goto start
:end
```

2 Math

2.1 Simpson

```
double a = 0, b = 1;
const int n = 100000000;
double s = 0;
double h = (b - a) / n;
fi(0, n) {
    double x = a + h * i;
    s += fun(x) * ((i==0 || i==n) ? 1 : ((i&1)==0) ? 2 : 4);
}
s *= h / 3;
```

3 Strings